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OCI-Dedicated KMS B&R user experience

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Introduction:

The DKMS B&R feature can be used via command line from customer VM where user management utility resides.

The backup will consist of

- Keys
- Users
- Config

Prerequisites:

- As a CO user the customer should create a POKBK and upload it to all the partitions of the cluster.
- The customer verifies that the POKBK is uploaded to all partitions.
- The customer can list the existing POKBK.
- The customer should back up the POKBK in an encrypted format.

POKBK (PCO-fixed.key)	Partition Owner Key Backup Key. The customer needs to upload this to a partition to perform backup and restore functions

On-demand cluster backup:

Customer initiates a cluster backup using a single command via the on the VM.

- No write operations are allowed during backup.
- DKMS checks if the same POKBK exists on all partitions.
- The backup succeeds if the POKBK exists in the partitions and the partition is up and running.
- Backup succeeds only when all partitions of the cluster are backed up successfully.
- In any other case, the customer gets an error message.
- The customer will need to provide the backup location (It can be VM or object storage bucket).

Restoring backup to a new cluster:

Customer will create a new cluster.

Using a single command, restore the cluster backup to a new cluster.

- The customer gets the backup bundle on the VM or provides the bucket location.
- Customer uploads the POKBK to the partition.
- Customer deletes the CU user if it exists on the partitions.
- Will this overwrite the content of the partition.

Restoring On-Demand Backups to the same cluster:

Currently this operation is not supported.

We would like to know if this is a requirement and what is the use case.

If supported:

Using a single command, restore all partitions of the cluster from the backup to the same cluster.

- The customer gets the backup bundle on the VM or provides the bucket location.
- Customer deletes the CU user if it exists on the partition.
- Will this overwrite the content of the partition.

Listing backups:

The customer can list backup using a list command

- The customer will provide the location on the VM/bucket where backups reside.
  - The response should provide:
    - The time when the back was taken
    - Key fingerprint for POKBK associated with the backup
    - **Any other metadata required?**

Key Checksum Value (KCV):	A checksum of a cryptographic key used to verify the key's integrity or to compare keys without revealing the key itself.

Thank you